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Spatially offset Raman scattering line-mapping as an adaptive tool ensuring accuracy for determination of component concentrations in tablets with different particle sizes

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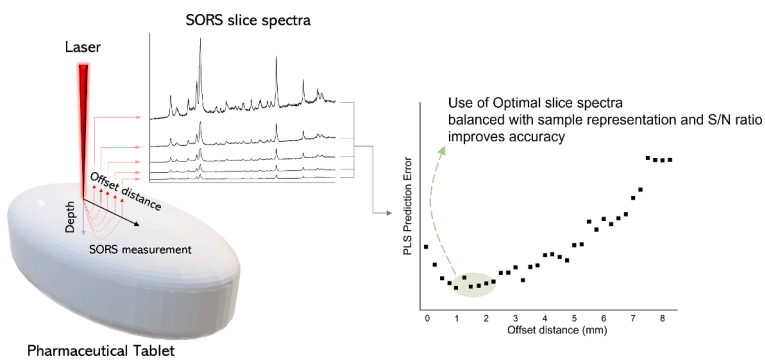
HIGHLIGHTS

- Spatially offset Raman scattering (SORS) was adopted to examine accuracy variation in tablet composition analyses.
- Thirty tablets with two different particle sizes were measured by SORS line-mapping.
- A series of slice spectra balanced with the sample representation and signal-to-noise ratio was existed.
- The use optimal slice spectra helped to improve accuracy for composition analysis of tablets.
- The offset distance-dependent accuracy variations differed according to particle size of tablets.

GRAPHICAL ABSTRACT



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ABSTRACT

Spatially offset Raman scattering (SORS) line-mapping was explored as a versatile tool to examine accuracy variations in compositional analyses of tablets with different particle sizes. SORS spectra collected near the laser irradiation were less representative of tablet composition due to the limited spectroscopic sampling volume, while the signal-to-noise (S/N) ratios of corresponding spectra were higher. On the other hand, SORS spectra at longer offset distances were better representative of tablet composition, while their S/N ratios were decreased considerably. Therefore, the use of only a certain portion of sliced (line-mapped) spectra balanced with the sample representation and S/N ratio could be advantageous to enhance accuracy. Moreover, a group of optimal slice spectra is expected to vary when the particle size of the tablet changes since the characteristics of internal photon propagation also would change. For the overall examination, SORS spectra of 30 Anaprox tablets (composed of 4 constituents including naproxen sodium) with 2 particle sizes (88.4 ± 11.8 μm and 118.9 ± 38.8 μm) were analyzed, and the concentrations of three components in these tablets were determined. A total of 6 cases (3 components and 2 particle sizes) were examined. When the average optimal slice spectra were employed in each case, the errors were lower compared to those using the average of all slice spectra. The demonstrated

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scheme was versatile to study the offset distance-dependent accuracy variations according to particle size and target component.

1. Introduction

Determination of active pharmaceutical ingredient (API) concentration in a tablet using Raman spectroscopy is a widely applied practice in pharmaceutical analysis since it is fast and non-destructive without need for sample pretreatment [1–4]. Typically, a back-scattering spectral acquisition scheme is adopted for tablet measurements due to the simple optical configuration and high efficiency of Raman photon collection. Various types of tablets targeting different afflictions are produced and need to be analyzed. Physical properties (such as particle size and packing density) can vary by tablet type but tend to be relatively consistent for a given tablet type. Under such circumstances, the propagation of laser photons and subsequent distribution of Raman photons vary by tablet type depending on their physical properties [5–7]. For example, Raman photon distributions became broader with increased particle size, and the associated variation directly influenced the accuracy of compositional analysis [8–10]. Therefore, a fixed spectral acquisition configuration would not be adequate (or optimal) for analyses of different types of tablets with dissimilar physical properties, and a measurement scheme should be individually adaptable to yield improved accuracy for a given tablet analysis.

In this context, spatially offset Raman scattering (SORS)-based line-mapping [11] of a tablet could be a viable choice to study the potential accuracy variations. Since the position of photon detection is physically separated from that of laser excitation in SORS line-mapping, a series of sliced (line-mapped) spectra with varying sample-interaction depths can be acquired. SORS spectra collected near the laser irradiation would be less representative of whole tablet composition due to the limited spectroscopic sampling volume near the tablet surface, although their signal-to-noise (S/N) ratios (intensities of peaks) are higher. On the other hand, SORS spectra at longer offset distances are better representative of whole tablet composition due to the increased sampling volume, though their S/N ratios considerably decrease due to the broadened Raman photon distribution (and fewer photons arriving at the detector). Therefore, the use of only a certain portion of slice spectra that is balanced in sample representation and S/N ratio could help ensure accuracy for compositional analysis. Moreover, the group of optimal slice spectra would vary when the physical characteristics of measured tablets are dissimilar. Thus, SORS line-mapping could be a viable tool to achieve improved accuracy by adaptively employing only optimal slice spectra for given tablet analyses.

Meanwhile, utilization of a SORS scheme has been expanded for detection of differences in bone matrix quality such as fracture resistance of bone [12] and quantitative fat analysis of milk through carton packing [13]. Also, chemometric algorithms such as multivariate curve resolution (MCR) and independent components analysis (ICA) were evaluated to enhance blind signal resolution and to highlight sample features from SORS spectra [14]. Moreover, SORS was cooperatively combined with near-infrared (NIR) spectroscopy to detect raw material defects in hazelnuts (*Corylus avellana* L.) [15].

To explore the potential of the suggested scheme, SORS line-mapping spectra were acquired for 30 Anaprox tablets (composed of naproxen sodium and 3 excipients) with 2 particle sizes in this study. For the 2 sample sets, the variations in peak intensity of naproxen sodium in the slice spectra were initially examined according to the offset distance from the laser. Next, offset distances yielding enhanced accuracies in determining component concentrations were identified in each case and discussed the results according to particle size of the sample.

2. Experimental section

2.1. Preparation of tablets

Anaprox tablets with a biconvex shape (height: 3 mm, length: 15 mm, width: 8 mm) were prepared at Chong Kun Dang Pharmaceutical Co., Cheonan, Korea. A photograph of a representative tablet is shown in Fig. 1 (a). The tablets were composed of the 4 constituents; naproxen sodium (API), microcrystalline cellulose (MCC), lactose hydrate, and magnesium stearate. The concentrations of naproxen sodium, MCC, and lactose hydrate in the tablets varied in the ranges of 55.6 – 82.0, 11.0 – 30.7, and 5 – 11.8 wt%, respectively, while the concentration of magnesium stearate was constant at 2.0 wt%. Initially, mixed powders of the components at the target concentrations with a particle size of 88.4 ± 11.8 μm were prepared and used to produce Type I tablets. Powders of the same compositions with an increased particle size of 118.9 ± 38.8 μm were used to prepare Type II tablets. Based on a previous report [16], the average particle sizes of 73 commercial batches manufactured during the timeframe of 2017–2020 ranged from 100 to 160 μm approximately. Also, the particle fractions below 125 μm were relatively dominant in the batches, ranging from 30 to 65 %. Therefore, based on this report, the particles sizes of 88.4 ± 11.8 μm and 118.9 ± 38.8 μm were representative of those of commercial products.

2.2. SORS line-mapping of the tablets and determination of component concentrations

Line-mapping spectra were acquired using an in-house wide-depth SORS system that consisted of a polychromator (Acton SP2300, Princeton Instruments, NJ, USA), a large-area CCD camera (PIXIS 400BR, Princeton Instruments, NJ, USA), a 785 nm diode laser (LML-785.0, PD-LD, NJ, USA), and relevant optical components. For spectral acquisition, the laser (power: 200 mW) irradiated a tablet for 3 s, and resurfacing Raman photons below the laser spot were collected using two achromatic lenses. Remaining photons were allowed to pass through a long-pass filter (LL01-785–25, Semrock, NY, USA) and were projected onto the slit of the polychromator. Offset-dependent Raman photons were dispersed by a grating and were collected with a CCD at 1340 × 400 pixels. Of the 400 rows in the CCD, 340 were used to acquire SORS spectra. The calculated offset size for an image pixel was 25 μm. With this instrumentation, the covered range of physical offset distance was from 0.25 to 8.5 mm.

SORS spectra of each tablet were collected by accumulating 10 scans each with a laser exposure time of 3 s (total acquisition time: 30 s). For each tablet, triplicate spectra were collected by randomly illuminating three locations. The baselines of raw spectra were initially corrected using asymmetric least-squares (AsLS). The AsLS-treated spectra were normalized by dividing the intensities of each peak by the corresponding peak area calculated over the 1800–250 cm^{-1} range, and the normalized spectra were used to determine the concentrations of components in the tablet.

For concentration determination, 24 and 6 of the total 30 tablets were randomly chosen for training and test sets, respectively. Partial least squares (PLS) models were developed using the training samples, and component concentrations of the test samples were predicted to calculate corresponding accuracies. This procedure of random sample selection was repeated 100 times for cross-validation, and the mean standard error of calibration (MSEC) and mean standard error of prediction (MSEP) were calculated. AsLS, normalization of spectra, and PLS were performed using MATLAB R2017b (The MathWorks Inc., MA, USA).

3. Results and discussion

3.1. Spectral features of pure components and SORS-line mapping spectra of tablets

Fig. 1 (b) shows the Raman spectra of pure naproxen sodium, MCC, lactose hydrate, and magnesium stearate powders in the 1800–250 cm^{-1} range. Diverse peaks of naproxen sodium with strong and weak intensities appear over the spectral range. Various peaks of the other excipients (components) are also present and overlapped with those of naproxen sodium. The peak intensities of the excipients are generally weaker than those of naproxen sodium. The naproxen sodium peak at 1380 cm^{-1} is most intense and moderately overlapped with the peaks of MCC and lactose hydrate. The second strongest peak appeared at 742 cm^{-1} but showed less overlap with those of the excipients. The spectra of a tablet with 68.75 wt% naproxen sodium, 20 wt% MCC, 9.25 wt% lactose hydrate, and 2 wt% magnesium stearate are shown in Fig. 1 (b). The overall peak shape of a tablet is quite similar to that of naproxen sodium due to the high content of naproxen sodium and its intense peaks. Determination of naproxen sodium concentration is expected to be not difficult as long as the acquired Raman spectra are representative of whole tablet composition with acceptable S/N ratios. The 1300–800 cm^{-1} range, which excludes the intense naproxen sodium peaks, is preferable for determination of MCC and lactose hydrate concentrations.

A total of 340 SORS spectra was collected from each tablet up to an offset distance of 8.5 mm. Among them, 10 consecutive slice (line-mapped) spectra from the first (for example, the 1st – 10th slice spectra) group were averaged into one slice spectrum. So, a total of 34 slice spectra obtained from each of the tablets was employed for subsequent analyses. Fig. 1 (c) shows all 34 slice spectra measured from the same tablet (Type I tablet, particle size: 88.4 ± 11.8 μm). The offset distances corresponding to the slice spectra are indicated at the right. Each of the slice spectra is properly offset for clarity. The top and bottom spectra are of the 1st and 34th slices, respectively. The intensities of tablet peaks including the main naproxen sodium peaks continuously decreased with increasing offset distance as expected since fewer photons arrived at the detector due to the greater radial diffusion of photons.

Fig. 2 (a) shows the variation of peak intensities at 742 cm^{-1} (I_{742}, black squares) in the 34 acquired slice spectra. The corresponding offset distances are designated on the upper *x*-axis. As seen, decreased peak intensities are predominant in the earlier slice spectra and the decrease pattern is exponential. The error bars at the corresponding offset distances were based on triplicate measurements as explained above. The average relative standard deviation (RSD) of intensities at the tested offset distances was 6.6 %. The S/N ratios for each slice spectrum are presented in Fig. 2 (b). For the calculation, the signal was set as the I_{742} value, and the noise was calculated as the standard deviation of intensities over the 1800–1700 cm^{-1} range, where tablets peaks were absent. As expected, the S/N ratios decreased somewhat linearly from the 1st to 34th slice spectra. The S/N ratios of the 20th to 34th slice spectra are less than 40 due to the lower intensities of tablet peaks.

For a detailed examination of the slice spectra at a comparable scale, the normalized 1st, 5th, 10th, 15th, 25th, 30th, ands 34th slice spectra are displayed in Fig. 3 (a). The spectra are vertically offset for a clear comparison, and the corresponding offset distances are designated at the right. The overall shapes of the spectra are analogous to each other, while the noise is considerably more noticeable in the latter slice spectra (e.g., 25th, 30th, and 34th). Therefore, the predicted component concentration is expected to be less accurate when the latter slice spectra are adopted. In addition, the earlier slice spectra of small sample-interaction volumes may not be sufficiently representative of whole tablet composition, which is known as the sub-sampling problem [17–19]. Therefore, the employment of certain slice spectra with balance between sample representation and S/N ratio can improve accuracy compared to using all slice spectra.

3.2. Determination of naproxen sodium concentration in Type I tablets

Naproxen sodium concentrations were determined using the slice spectra of all Type I tablets acquired at each offset distance with PLS. Fig. 4 (a) presents the MSEPs when the first through 34th slice spectra are separately employed for concentration determination. In all cases, four factors were adopted for each PLS model. Interestingly, the MSEP-1st (MSEP using the first slice spectra) was higher than the subsequent MSEPs such as MSEP-2nd and MSEP-3rd, although the peak intensities in the first slice spectra were strongest (the highest S/N ratio). The decrease in MSEP continued until the fifth slice spectra. This suggests that the spectroscopic sampling volume has greater impact on accuracy than does the S/N ratio in this short offset distance range. In other words, the first slice spectra are not sufficiently representative of the

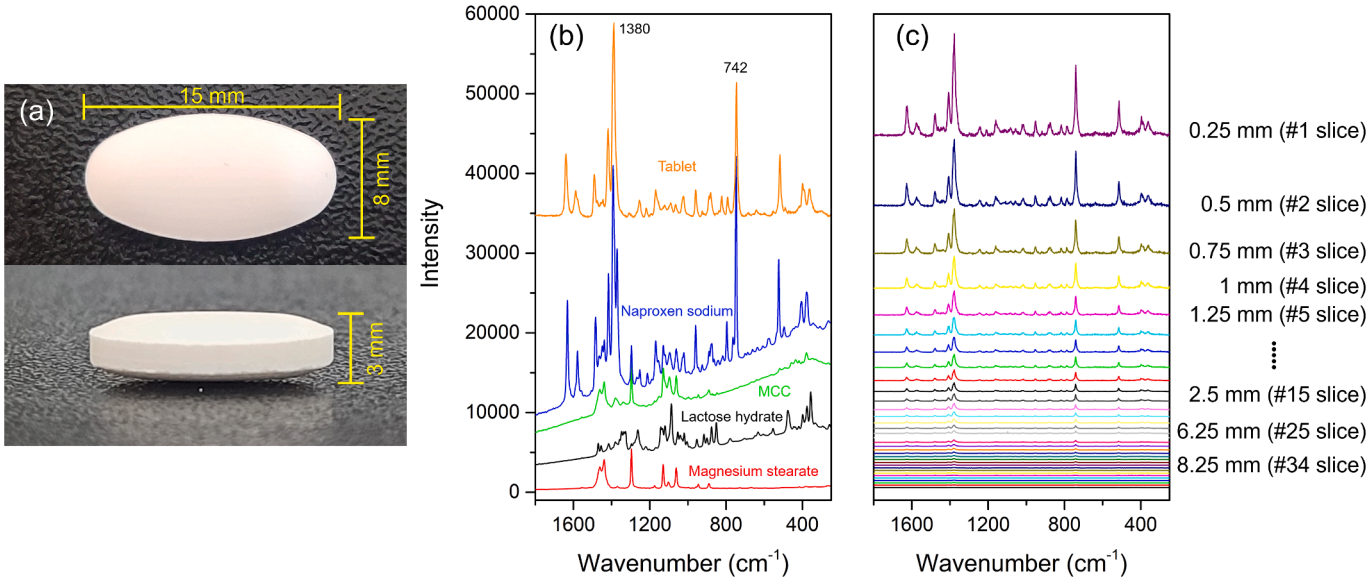

Fig. 1. Pictures of a measured tablet (a), Raman spectra of pure naproxen sodium, MCC, lactose hydrate, magnesium stearate, and the Type I tablet containing 68.75 wt% naproxen sodium (refer to the corresponding pictures inside) acquired with a conventional back-scattering mode (b), and all 34 slice spectra from the same tablet measured by SORS line-mapping (c). Slice spectra are offset for clarity.

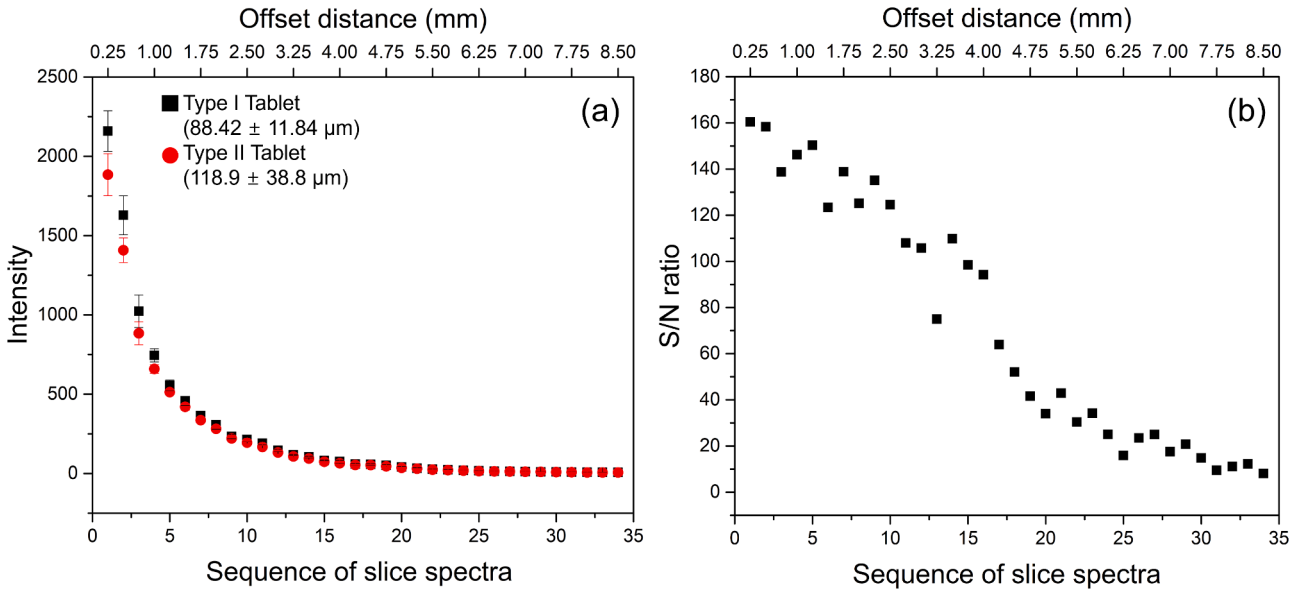

Fig. 2. Variations of I_{742} values in the 34 slice spectra acquired from Type 1 (black square) and Type II (red circle) tablets containing 68.75 wt% naproxen sodium (a) and S/N ratios of the 34 slice spectra of a Type I tablet (b).

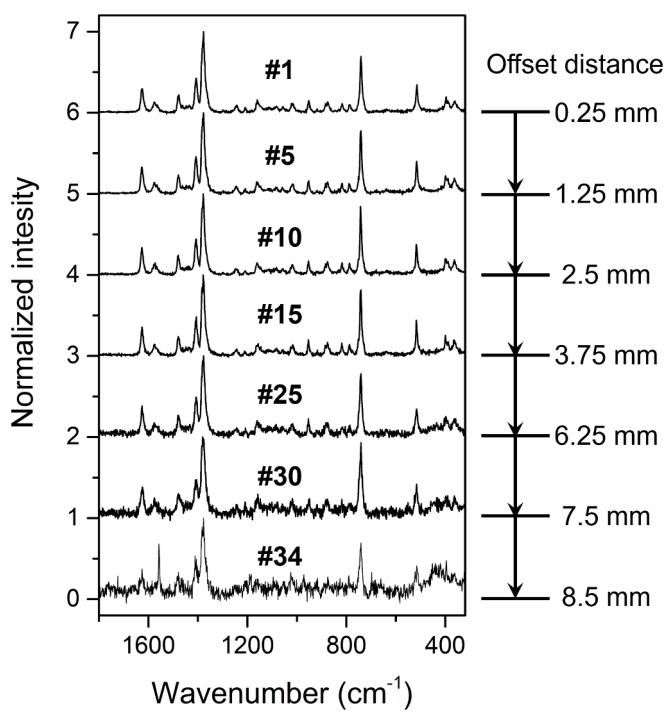

Fig. 3. Normalized 1st, 5th, 10th, 15th, 25th, 30th, and 34th slice spectra of the Type I tablet containing 68.75 wt% naproxen sodium. The offset distances corresponding to each slice spectrum are designated at the right.

whole tablet composition because they have the smallest sampling volume from the shortest offset distance of 0.25 mm. The fifth slice spectra acquired at the longer offset distance of 1.25 mm yielded a greater sampling volume and are better representative of the whole tablet composition, with a decrease to 0.88 wt% for MSEP-5th from 1.56 wt% for MSEP-1st.

The MSEP-5th through MSEP-10th were maintained at lower levels around 0.9 wt%. This suggests that the corresponding offset distances (1.25——2.5 mm) allowed production of slice spectra more representative of whole tablet composition as well as a good S/N ratio. The remaining MESPs after MSEP-11th increased although the corresponding spectra were sufficiently representative of the tablet composition. The S/N ratios of spectra start to govern the accuracies in this offset distance range (from 2.5 to 8.5 mm). These observations confirmed the presence of optimal slice spectra that yield the highest accuracy among all slice spectra.

The MSEP was 0.99 wt% when averaging all slice spectra of each tablet, and these average spectra were used for PLS. In contrast, the MSEP decreased to 0.79 wt% when averaging only the 5th, 6th, 7th, 8th, 9th, and 10th slice spectra. The corresponding concentration correlation plot is shown in Fig. 5 (a). Rather than using only the 5th slice spectra, the average of these optimal slice spectra enhanced the S/N ratio and helped to improve the robustness of concentration prediction. So, the use of the averaged spectra of optimal slice spectra balanced with the sample representation and S/N ratio helped to enhance the accuracy.

To examine the differences among the earlier slice spectra (1st to 9th), the difference spectra obtained by subtracting each of the normalized slice spectra from the #5 normalized slice spectrum (designated as #1–5, #2–5, #3–5, #4–5, #6–5, #7–5, #8–5, and #9–5 difference spectra) were examined as shown in Fig. 6. In all the spectra, the intensity differences are observed around 1628, 1379, and 742 cm^{-1}, corresponding to the main naproxen sodium peaks, and the magnitudes of differences are small. Meanwhile, the magnitudes of intensity differences are relatively high for the #1–5, #2–5, #3–5, #7–5, #8–5, and #9–5 difference spectra, and low for the #4–5 and #6–5 difference spectra. This observation confirms that the compositional representations of the consecutive slice spectra is dissimilar to each other due to the different sampling volumes at the relevant offset distances.

3.3. Determination of naproxen sodium concentration in Type II tablets

As previously reported, the intensities of Raman peaks of a measured solid sample and their reproducibility of measurement vary by particle size [20]. Therefore, when tablets composed of particles with different sizes are measured using the same SORS-line mapping, dissimilar offset distance-dependent behavior of MSEPs will be observed. For examination, SORS line-mapping was performed for Type II tablets with larger particle size (average particle size: 118.9 ± 38.8 μm). Fig. 2 (a) presents the I_{742} values (red circles) in the 34 acquired slice spectra of Type II tablets with the same composition as but larger particle size than Type I

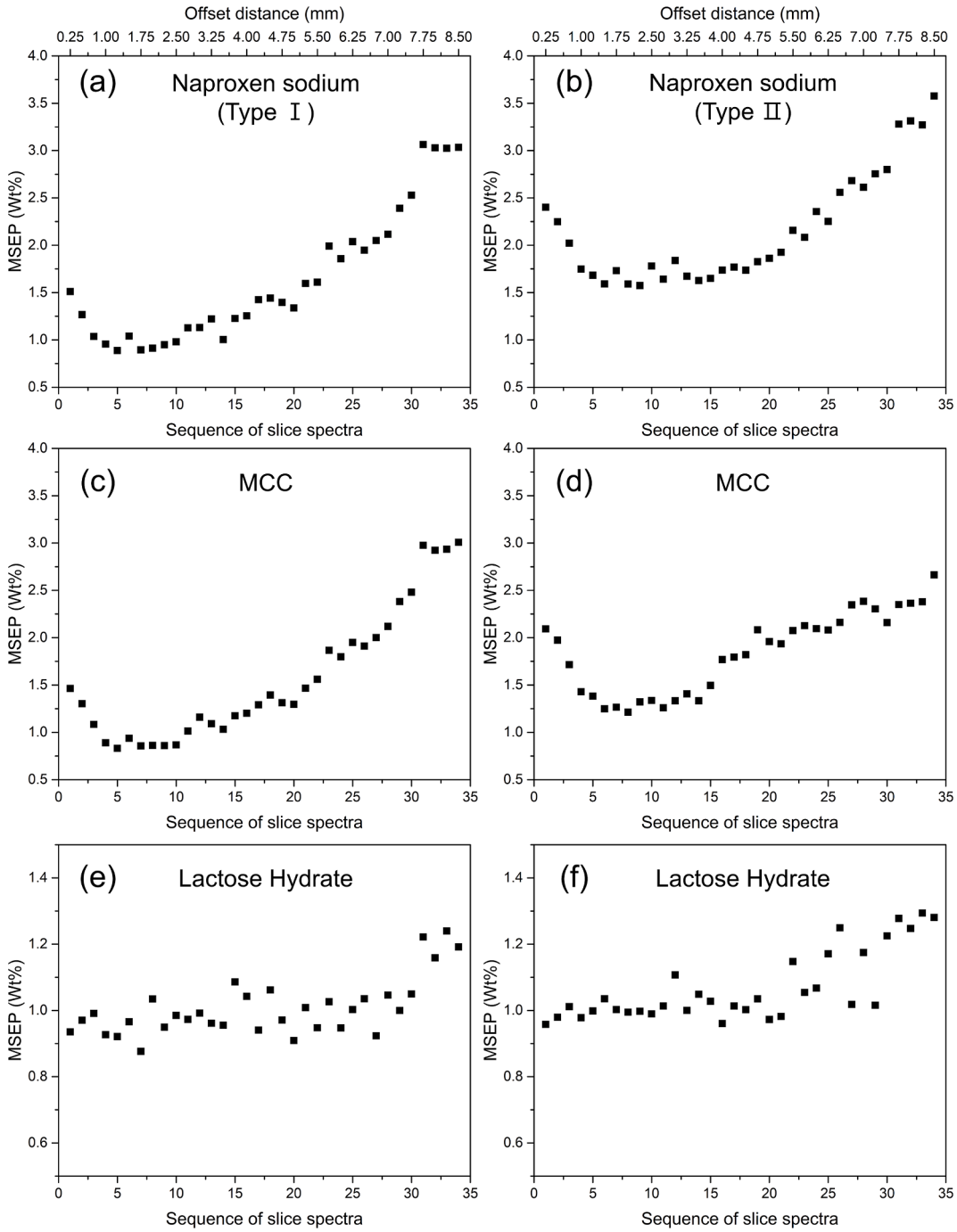

Fig. 4. Variations of MSEPs when the 1st through 34th slice spectra of Type I (left column) and Type II (right column) tablets are individually employed for the determination of naproxen sodium ((a), (b)), MCC ((c), (d)), and lactose hydrate ((e), (f)) concentrations using PLS.

tablets. As shown, the I_{742} values of Type II tablets at each offset distance are all lower than the respective values of Type I tablets. The average RSD of intensities at the tested offset distances was 5.6 %. The differences in I_{742} values between the two types of tablets were greater at the earlier offset distances, from 0.25 to 1.25 mm (producing the 1st to 5th slice spectra).

As reported in a previous publication [11], when a sample composed of large particles is measured using Raman spectroscopy, the laser diffuses over a longer distance inside the sample due to the longer mean free path caused by the larger particle size. This results in a broader distribution of Raman photons. As the size of the pixel row in the CCD was fixed in the present study, the number of Raman photons arriving at the pixel decreased due to the broadened photon distribution. This explains the lower peak intensities of the Type II tablets.

Fig. 4 (b) presents the MSEPs when the 1st through 34th slice spectra of Type II tablets were separately employed for concentration determination. Notably, the MSEPs at each offset distance are all higher than the respective MSEPs in the measurement of Type I tablets (Fig. 4 (a)). Since fewer Raman photons that interact with the specimen arrive at each pixel in the measurement of Type II tablets, each slice spectrum is less representative of whole tablet composition and produces a less accurate compositional analysis of tablets. Compared to Type I tablets, the slice spectra yield lower MSEPs, such as those of the 6th to the 14th slice spectra. This suggests that MSEP variation according to offset distance

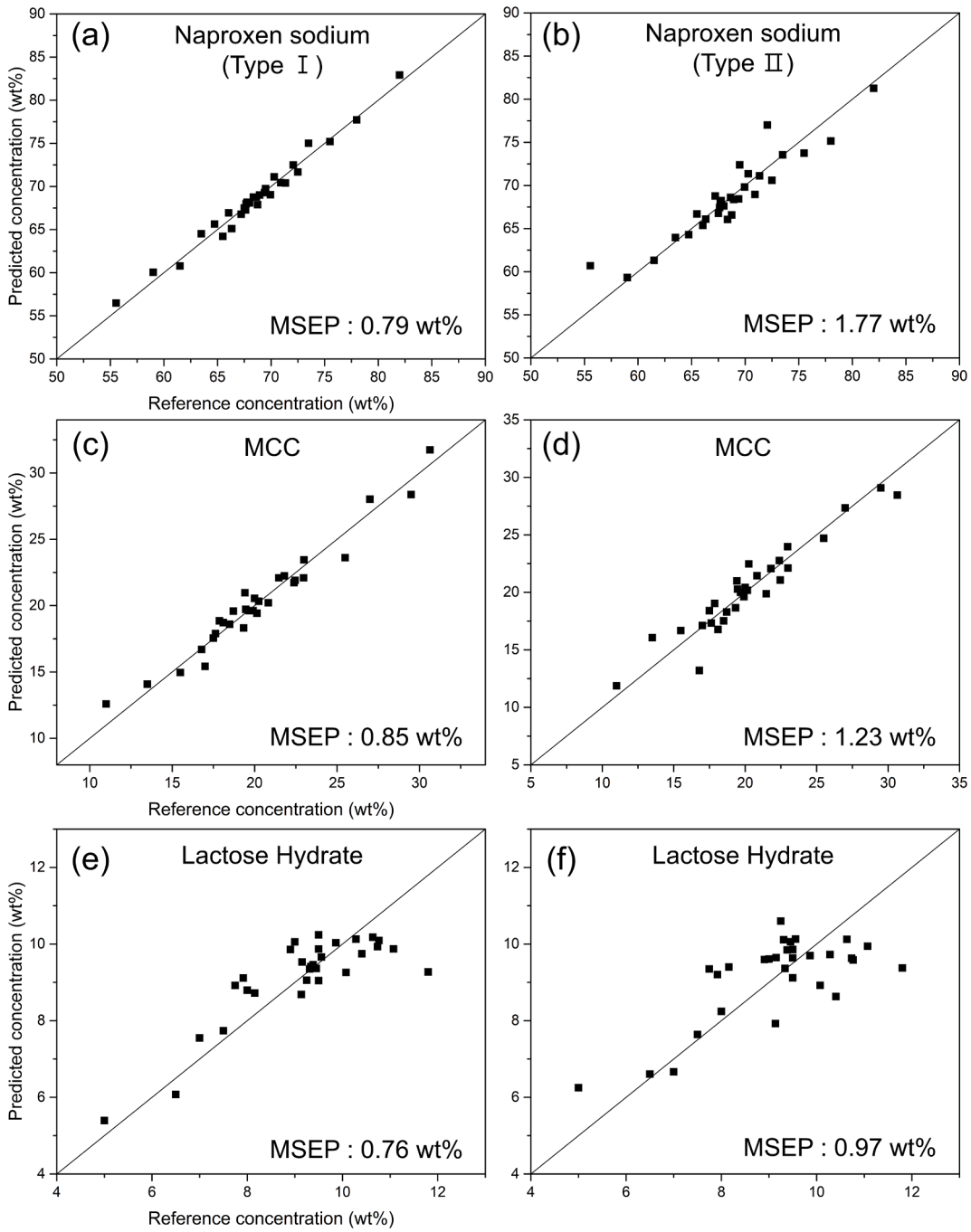

Fig. 5. Correlation plots of naproxen sodium ((a), (b)), MCC ((c), (d)), and lactose hydrate ((e), (f)) concentrations of Type I (left column) and Type II (right column) tablets when the average optimal slice spectra in each case were used for PLS.

varies by particle size of a tablet.

When the average spectra of all slice spectra collected from each of the Type II tablets were used for PLS, the MSEP was 2.01 wt%, higher than the corresponding MSEP (0.99 wt%) of Type I tablets. Similarly, the MSEP decreased to 1.77 wt% when using average spectra of the 6th through 14th slices. The corresponding concentration correlation plot is shown in Fig. 6 (b). For Type II tablet measurements, employment of optimal slice spectra improved the accuracy, and the degree of accuracy enhancement was greater than that in the Type I tablets. This demonstrates that the employed SORS-line mapping is an effective tool to adaptively secure accuracy for compositional analysis of tablets even when their particle sizes vary.

3.4. Determination of MCC and lactose hydrate concentrations in Type I and II tablets

Since the concentrations and peak intensities of MCC and lactose hydrate in the tablets are lower than those of naproxen sodium (refer to Fig. 1 (b)), it is interesting to examine their offset distance-dependent MSEP variations. Since the strong peaks of naproxen sodium are present in the 1650–1300 and 780–450 cm^{-1} ranges, they likely hamper recognition of low-intensity MCC and lactose hydrate peaks and should be excluded for the concentration determination of these two components. Therefore, the 1300–800 cm^{-1} range, which did not contain the two strongest naproxen sodium peaks, was employed for PLS.

Fig. 4 (c) and (d) show the variations of MSEPs according to offset

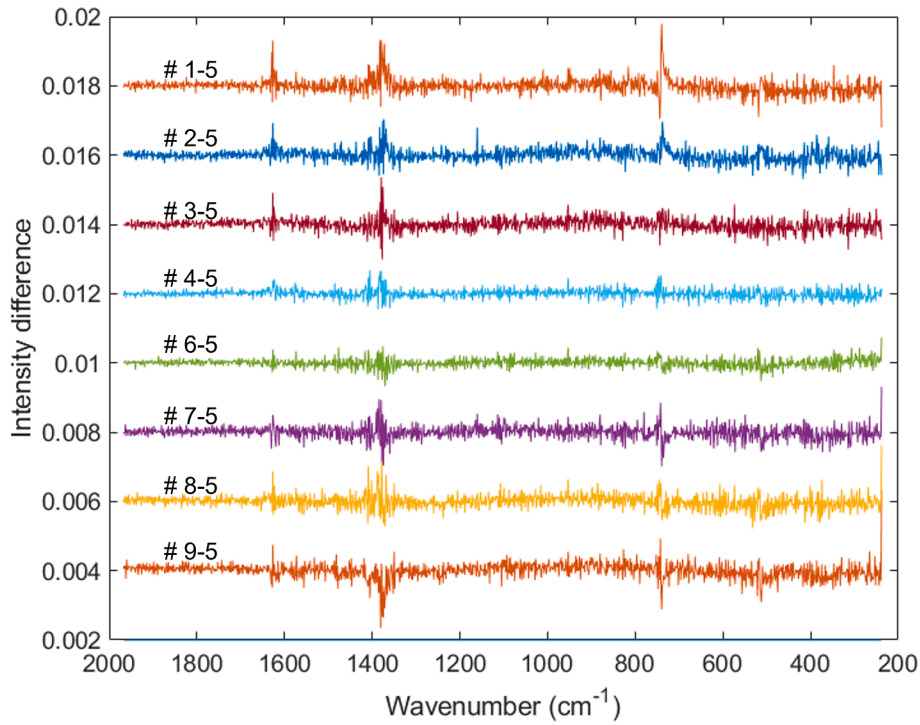

Fig. 6. Difference spectra obtained by subtracting each of the normalized slice spectra from #5 normalized slice spectrum (designated as #1–5, #2–5, #3–5, #4–5, #6–5, #7–5, #8–5, and #9–5 difference spectra).

distance in determination of MCC concentrations in Type I and II tablets, respectively. For Type I tablets, the pattern of MSEP variation is similar to that in the determination of naproxen sodium concentration (Fig. 4 (a)), such as the lower MSEP-5th through MSEP-10th. When the average spectra of all slice spectra from each tablet were used for PLS, the MSEP was 1.01 wt%, which decreased to 0.85 wt% when using the average spectra of the 5th through 10th slice spectra (refer to the corresponding concentration correlation plot in Fig. 5 (c)).

The patterns of MSEP variations in the determination of MCC (Fig. 4 (d)) and naproxen sodium (Fig. 4 (b)) concentrations in the measurement of Type II tablets are analogous. For example, they are lower for MSEP-6th through MSEP-14th. When the average spectra of all slice spectra was used for PLS, the MSEP was 2.11 wt%. However, the MSEP decreased to 1.23 wt% when employing the average spectra of 6th through 14th slice spectra (the corresponding concentration correlation plot in Fig. 5 (d)). Note that the MSEPs at each offset distance in the measurement of Type II tablets were all higher than the respective MSEPs in the measurement of Type I tablets.

Fig. 4 (e) and (f) display the variations in MSEPs according to offset distance in the determination of lactose hydrate concentrations in Type I and II tablets, respectively. The patterns are different from those of naproxen sodium and MCC, with greater MSEP scatter from the trend curve in both cases and gradually increased without significant minimal locations. When the average spectra of all slice spectra were used, the MSEP was 0.94 wt%, which decreased to 0.76 wt% using the average spectra of 4th through 7^{h} slice spectra (the corresponding concentration correlation plot in Fig. 5 (e)). In the measurement of Type II tablets, MSEP-1st through MSEP-10th are similar to each other, with an MSEP of 0.97 wt% (the concentration correlation plot in Fig. 5 (f)). When using the average spectra of all slice spectra, the MSEP was 0.99 wt% and was not significantly degraded.

4. Conclusions

The offset distances yielding the lower MSEPs differed depending on particle size and target component for concentration determination in this study. The demonstrated SORS line-mapping scheme is a simple and effective tool to identify the improved accuracy for each of the six studied cases (refer to Figs. 4 and 5) by adaptively selecting the slice spectra that balance the sample representation and S/N ratio for compositional analysis of tablets. Meanwhile, a broad-illumination Raman measurement also enabling of more representative spectroscopic sampling [21] has been frequently adopted for the compositional analysis of diverse tablets. Therefore, the performance comparison of SORS and broad-illumination schemes is an interesting issue in non-destructive tablet analysis, and will be executed as a future study. Other than particle size, diverse factors such as refractive index and absorption coefficient of particles, and packing density of tablets potentially influence photon propagation inside a tablet and curvature (shape) of a tablet is also a concern. Therefore, our group is preparing various tablets with different physical properties and shapes for this comparative study. Extensive Monte Carlo simulation with optimization of diverse photon-propagation parameters will be performed in parallel.

CRediT authorship contribution statement

Sanghoon Cho: Resources, Methodology, Data curation. **Jin Il Jang:** Formal analysis. **Hyung Min Kim:** Methodology, Data curation. **Jaejin Kim:** Supervision, Conceptualization. **Hoeil Chung:** Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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